

NAÏVE BAYES CLASSIFIER

Naïve Bayes is a probabilistic classifier based on Bayes' theorem. It assumes that all the attributes are independent of each other, each contributing to the probability of the class outcome. This independence assumption is not always exactly true in real data but Naïve Bayes gives good results on data sets with many categorical attributes (Witten et al., 2017). There are 19 input attributes in this dataset, 13 of them are categorical, such as Motivation_Level, Parental_Involvement, Access_to_Resources, Family_Income, Peer_Influence and School_Type, so Naïve Bayes is a natural and well-justified comparison classifier.

In WEKA, I used useSupervisedDiscretization to enable the algorithm to automatically determine the best intervals to bin the numeric attributes (Attendance and Hours_Studied) so that the probability calculations could be made with them. I used Cross-validation with 10 folds and checked that the class is (Nom) Exam_Score and then pressed Start.

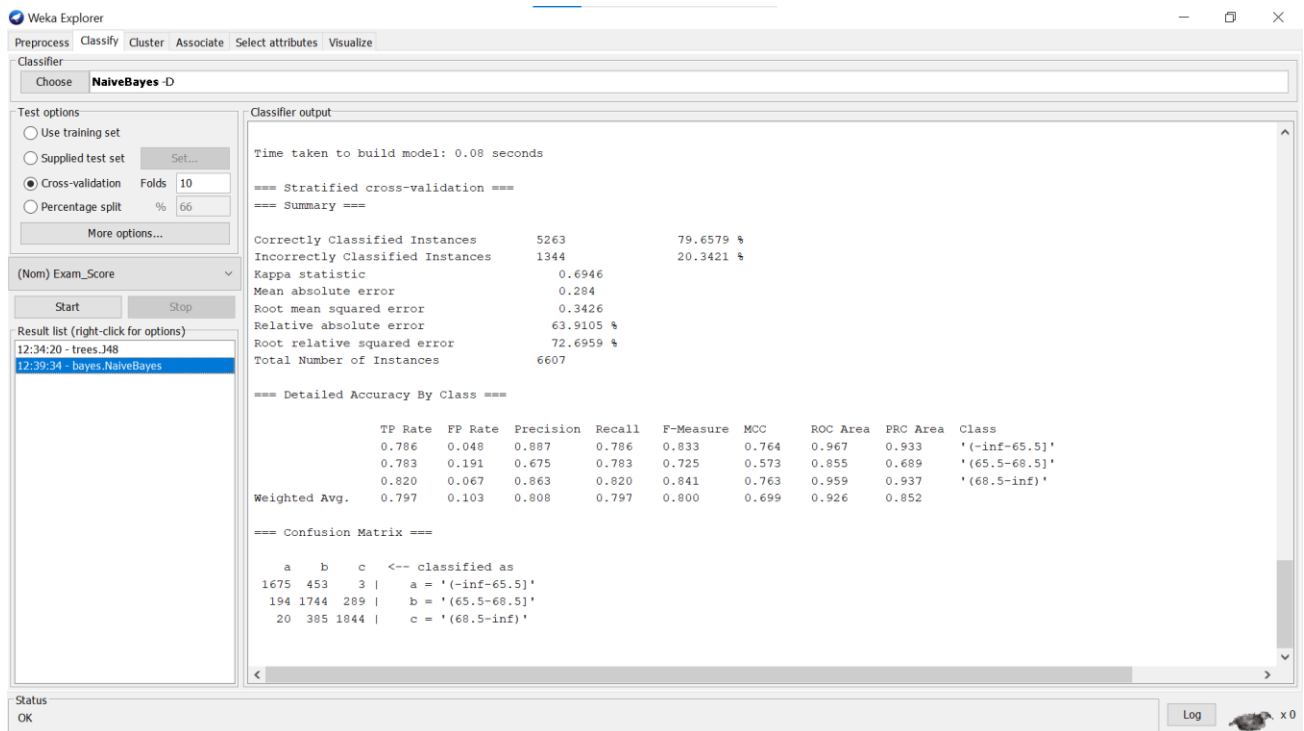


Figure 6 : WEKA Classify Tab with NaiveBayes -D configuration. 10-fold cross validation, class = (Nom) Exam_Score. Overall accuracy = 79.66%, Kappa statistic = 0.6946.